

Key Concepts in Macroeconomics

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus Le Havre

Lecture 1

Aggregate Output

Why we care about GDP?

When economists look at a country, the first two questions they ask are: How big is the country from an economic point of view? And what is its standard of living? Economic condition?

- Output: the level of production of the country as a whole
- Output per person
- Output growth: the rate of change of output
- The unemployment rate: the proportion of workers in the economy who are not employed and are looking for a job
- The inflation rate: the rate at which the average price of goods in the economy is increasing over time

Aggregate output: Motivation

Motivation: What is GDP? Why do we care?

- GDP is a measure of output
- We care about GDP because output is (often times) highly correlated with other things we care about (standard of living, wages, unemployment, inflation, value of currency, etc. . .)
- **Formal Definition:** GDP is the Market Value of all final goods and services newly produced on domestic soil during a given time period

GDP: Let's break this down

GDP is the **Market Value** of all final goods and services newly produced on domestic soil during a given time period

- We can add the production of different goods and services (G&S)
- G&S sold in formal markets (child-rearing vs. day-care)
- Underground economy not counted

GDP: Let's break this down

GDP is the Market Value of **all final goods and services** newly produced on domestic soil during a given time period

- Only final goods, not intermediate goods
- Intermediate G&S are those used up in the production of other G&S in the same period that they themselves are produced
- Many subtle distinctions, e.g. inventories are counted in the GDP

GDP: Let's break this down

GDP is the Market Value of all final goods and services **newly produced** on domestic soil during a given time period

- GDP is a measure of current economic activity
- House built this year – included in GDP
- Used house sold this year – not included in GDP

GDP: Let's break this down

GDP is the Market Value of all final goods and services newly produced on **domestic soil** during a given time period

- GDP vs GNP
- GNP: Market Value of all final goods and services newly produced by national factors of production (firms, workers etc.) during a given time period
- $GDP = GNP - \text{Net Factor Payments from Abroad (NFP)}$
- NFP: income paid to domestic factors of production by the rest of the world - income paid to foreign factors of production by domestic economy

GDP: Let's break this down

GDP is the Market Value of all final goods and services newly produced on domestic soil **during a given time period**

- GDP is a flow variable (expressed as quantity per period of time)
- Wealth is a stock variable (quantity at a point in time)

Measuring methods in practice

1. Production method: the sum of value added summed across industries in the economy during a given period

$$\text{GDP} = \text{sale prices} - \text{costs of raw materials}$$

2. Expenditure method: spending by all agents in the economy: consumers (C) + spending by businesses (I) + spending by government (G) + net Spending by foreign sector (NX)

$$\text{GDP} = C + I + G + NX$$

3. Income method: the sum of incomes in the economy during a given period

$$\text{GDP} = \text{Labor Income (wages/salary)} + \text{Capital Income (rent, interest, dividends, profits)}$$

Real GDP: Example

Compute the GDP:

	Steel Company (Firm 1)	Car Company (Firm 2)
Revenue for Sale	\$100	\$200
Expense (Cost)		
Wage	\$80	\$70
Intermediate good purchase	0	\$100 (steel)
Profit (Revenue - Cost)	\$20	\$30

Nominal and real GDP

- Nominal GDP is output valued at current prices.
- Comparing nominal GDPs over time can become problematic. Nominal GDP can increase for two very different reasons:
 1. The production of most goods increases
 2. The price of most goods increases
- Real GDP is output valued at some constant level of prices
 - Prices in a base year

Real GDP: Example

Compute the nominal and real GDP for the following economy:

	Year 1	Year 2
Product (quantity)		
Computers	5	10
Bicycles	200	250
Price (per item)		
Computers	\$1200	\$600
Bicycles	\$200	\$240

- Nominal GDP: Year 1 = \$46,000, Year 2 = \$66,000
- Real GDP (base year = year 1): Year 1 = \$46,000, Year 2 = \$62,000
- Real GDP (base year = year 2): Year 1 = \$51,000, Year 2 = \$66,000

Real GDP: Base years and chain-weighted GDP

- We have just seen that the choice of the base year matters
- Which base year is the right one?
 - There is no clear reason to prefer year 1 over year 2
- In 1996: the Bureau of Economic Analysis introduced chain-weighted indexes
 - Mathematical compromise
 - Chain-weighting effectively updates the base year automatically.
 - The annual growth rate for a given year is computed using that year and a preceding year as base years.

Construction of chain-weighted GDP

1. Construct real GDP using each year as base year

	Year 1	Year 2
base year = Y1	46,000	62,000
base year = Y2	51,000	66,000

2. Calculate GDP growth for each method

$$\Delta \text{GDP} = \frac{\text{GDP}_t - \text{GDP}_{t-1}}{\text{GDP}_{t-1}}$$

$$\Delta \text{GDP}_{\text{baseY1}} = \frac{62,000 - 46,000}{46,000} = 0.35 \quad \Delta \text{GDP}_{\text{baseY2}} = \frac{66,000 - 51,000}{51,000} = 0.29$$

3. Construct new GDP growth rate as the average of these growth rates

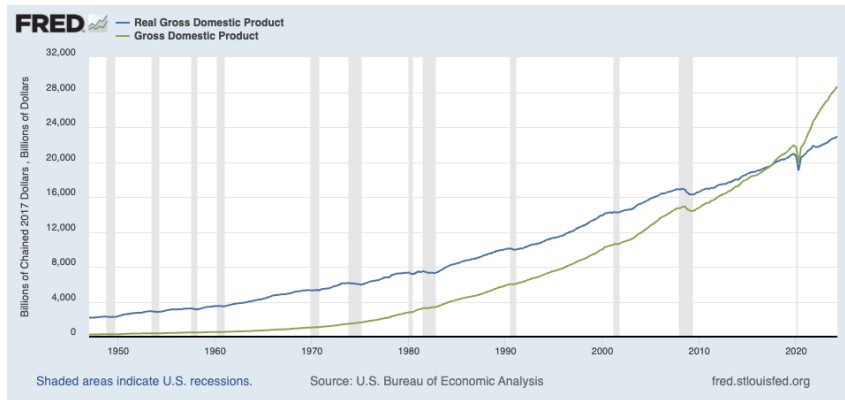
$$\frac{0.35 + 0.29}{2} = 0.32$$

Construction of chain-weighted GDP (cont'd)

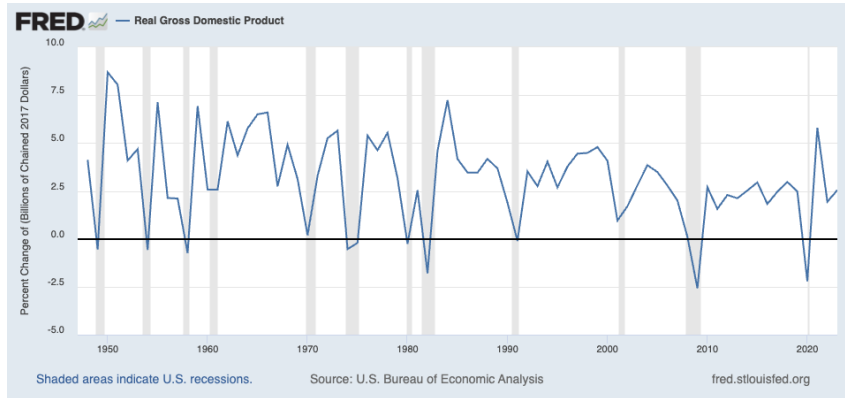
4. Select a base and build the chain

Year	Index	Chained GDP
Year 1	1	$1 \times 46,000 = 46,000$
Year 2	$1 + 0.32$	$1.32 \times 46,000 = 60,720$

Nominal and Real GDP, US (1960–today)



GDP growth and the business cycle



Unemployment rate in the US



Unemployment

Unemployment: Key concepts

- **Employment (E):** the number of people who have a job
- **Unemployment (U):** the number of people who do not have a job but are looking for one
- **Labor force (L):** the sum of employment and unemployment

$$L = E + U$$

- **Unemployment rate (u):** a ratio of the number of people who are unemployed to the number of people in the labor force:

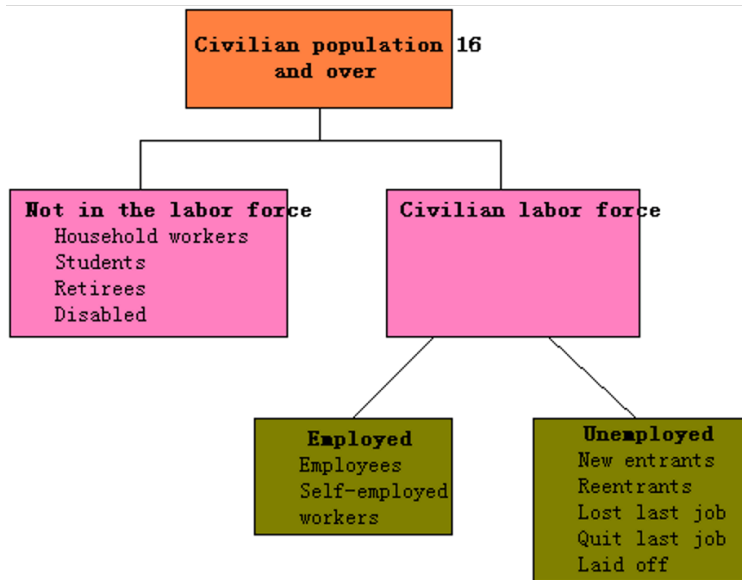
$$u = \frac{U}{L}$$

This is the definition used in most countries, including the US

Unemployment: Measurement challenges

- Measurement issues (survey data)
- Underemployment
- Discouraged workers
 - A person is unemployed if he or she does not have a job and has been looking for a job in the last four weeks.

Definition of Unemployment



Unemployment: A Few More Definitions

Labor Force Participation Rate:

$$\frac{L}{Population(age16+)}$$

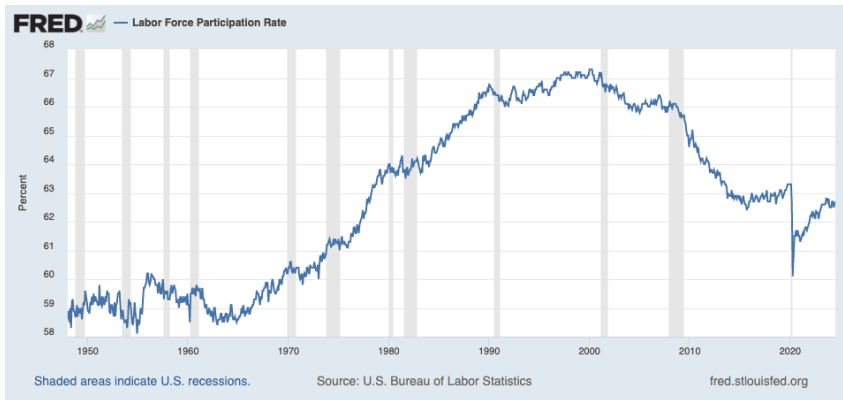
Employment Rate:

$$\frac{E}{Population(age16+)}$$

Unemployment rate in the US



Labor force participation rate in the US



Take aways

- Unemployment rate can be giving a misleading statistic of the US labor market
- Unemployment rate is low by historical standards
- Workers have left the labor force and are no longer looking for jobs

Inflation

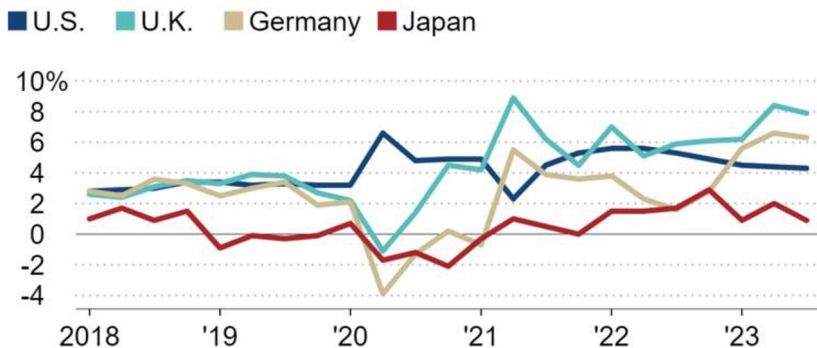
Key definitions

- **Inflation** is a sustained rise in the general level of prices
- **Inflation rate** is the percentage increase in the average level of prices over some period (often a year)
- **Deflation** is a sustained decline in the price level

Why Do Economists Care about Inflation?

- Pure inflation is proportional increase in all prices and wages
 - Only a minor inconvenience – as relative prices are unaffected.
 - Real wage (wage measured by goods rather than dollars) would be unaffected.
 - There is no such thing as pure inflation.
- Inflation affects income distribution when not all prices and wages rise proportionally.
- Inflation leads to distortions due to uncertainty, some prices that are fixed by law or by regulation, and its interaction with taxation.

Inflation and Real Wage Growth

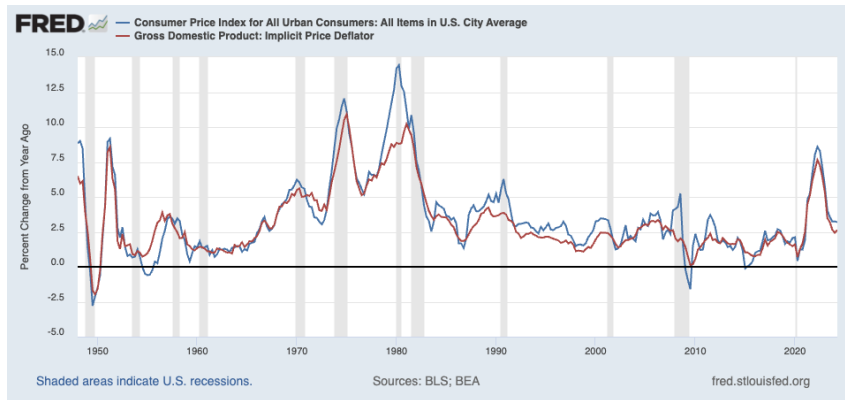


Source: Japan's Cabinet Office

Price Indexes

- Price index is a measure of the average level of prices for some specified set of goods and services, relative to the prices in a specified year.
- **GDP deflator** is a price index that measures the overall level of prices of goods and services included in the GDP
- **Consumer price index (CPI)** is a price index that measures the overall level of consumer goods

Price Indexes: CPI vs GDP deflator



GDP deflator

- The GDP deflator in year t is the ratio of nominal GDP to real GDP in year t

$$P_t = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

- It is called an index number P_t (1 in 2012), which has no economic interpretation.
- The rate of change has a clear interpretation: the rate of inflation.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

GDP deflator and GDP growth

- Defining the price level as the GDP deflator implies a simple relation between nominal GDP, real GDP, and the GDP deflator:

$$\text{Nominal GDP} = \text{Real GDP} \times P_t$$

- The rate of growth of nominal GDP is equal to the rate of inflation plus the rate of growth of real GDP.

Consumer Price Index (CPI)

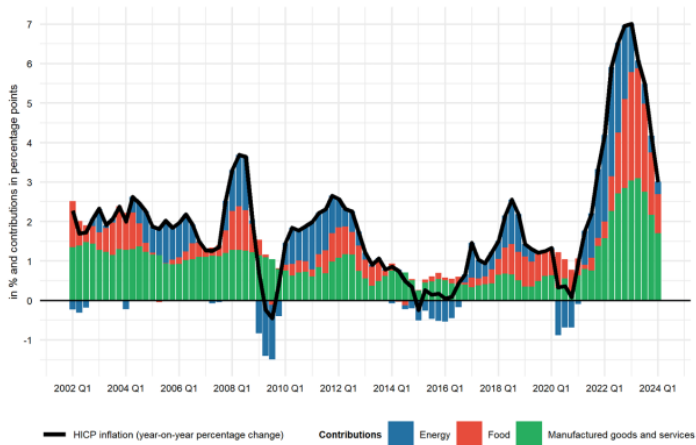
- CPI is a measure of the cost of living.
- CPI gives the cost in dollars of a specific basket of G&S over time.

Why are CPI and GDP deflator different?

G&S produced in the economy $\neq$ G&S purchased by consumers

- Some G&S are sold not to consumers but to firms, government or foreigners
- Some G&S bought by consumers are not produced domestically (imports)

What causes CPI to increase?



Data source: Bnque de France

Cost-push Inflation

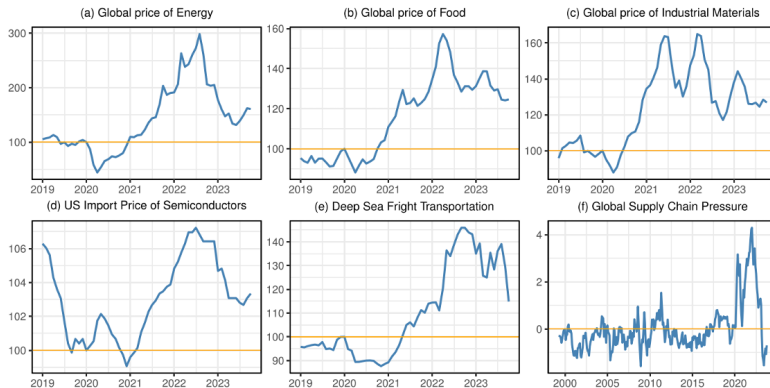


Figure 1. The post-pandemic global price shock: indices for selected categories of goods and services. *Note:* For panels (a)–(e): 2020=100. (Sources: IMF, BLS, Federal Reserve Bank of New York)

Data source: : <https://doi.org/10.1093/icc/dtae003>

Seller's inflation

- Does the recent inflation as a cost-push-profit-led inflation? Isabella Weber
- Seller's Inflation
- Weber et al (2025)
 - large input price shocks (as well as their co-occurrence with supply constraints) correlate with positive sentiments expressed in executives' statements about cost increases.
 - There are executives' optimism regarding their ability to turn an economy-wide cost shock into an opportunity to raise prices and protect or even increase profits.

Weber et al (2025)

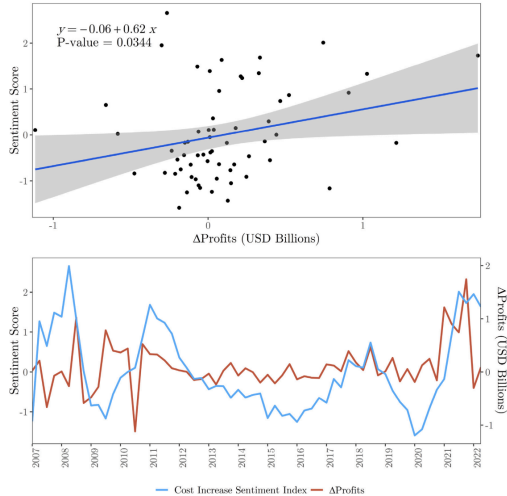


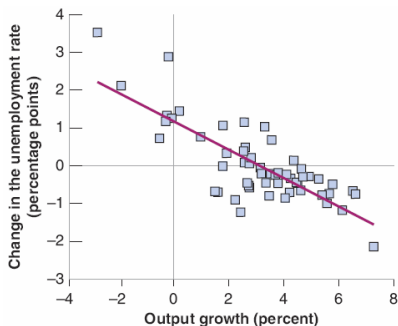
Figure: Indication between profit increase and cost increase sentiment

Output, unemployment and inflation

Okun's law

- Intuition: High output growth $\implies$ decrease in unemployment
- Okun's law: a rule of thumb, quantitative impact
- First described by the U.S. economist Arthur Okun

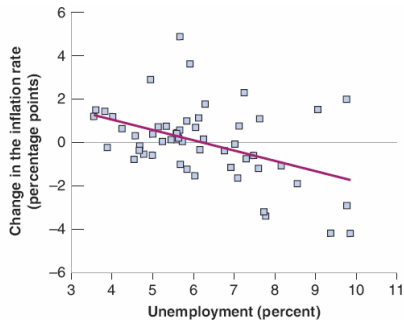
$$\Delta Y = 0.3 - 0.4\Delta u$$



- An increase in the growth rate of 1% decreases the unemployment rate by -0.4% .
- 3% growth rate of output keeps the unemployment constant

Phillips curve

- Intuition: when unemployment becomes very low $\implies$ the economy is likely to overheat $\implies$ high inflation
- Phillips curve illustrates this relationship empirically



- The line is downward sloping
- Higher unemployment leads, on average, to a decrease in inflation, and vice versa.